

NMAP

A Re-cap

Part—8

My first article on NMap, published in August 2010, covered just the basics. Later, the LFY team and a few knowledgeable readers encouraged me to cover NMap in detail, leading to this series of articles, which gave an overview of NMap, covering various aspects of it, and its practical use in network scanning. Some live scans were also included for a better understanding. This final article in the series consolidates information from the previous articles, looks at NMap's future possibilities, and at the new tools from the development team.

Beginning its life as a simple port-scanner, NMap has evolved into an excellent network security auditing tool. The NMap website now describes it as a free and open source utility for network exploration or security auditing.

Like most open source utilities, NMap is released under the GNU GPL license (free to use, modify, and distribute). Interested users can download the latest version for their OS and start using it. All versions have the same command-line syntax and the same GUI; the only differentiator is a person's knowledge of how to use it to scan/audit the network. Understanding the various command-line options should help you use the tool in the most effective way. For example, while demonstrating the NMap Scripting Engine capabilities at Black Hat 2010, the

following steps were performed live:

1. Tracking a live web-cam installed on an unknown public IP.
2. Brute-forcing its username/password to gain access.
3. Watching the live video displayed.

The seemingly impossible feat was performed in less than 15 minutes. Sure, it is definitely not easy to achieve this kind of expertise, but to effectively master NMap for every-day use should not be too difficult.

NMap command-line options

If you run *nmap* without any switches, it gives you a list of all available command line options. These are logically classified as shown in Table 1.

Table 1: A summary of NMap commands

Scan switch	Utilisation
Target specification	You can specify the target in various intuitive ways: by directly specifying the host-name/IP address; or by giving the start and end addresses to specify a range. You can also pass a list of IP addresses to NMap using the <code>-iL</code> switch, followed by the name of the file containing the IP list. You may also exclude hosts with the <code>--exclude <hostname(s)></code> switch.
Host discovery	From a wide range of hosts to be scanned, you will probably be interested in finding specific hosts, depending on the reason for the scan. NMap has various host-discovery techniques; some of the important ones are: -sL which will list the hosts to be scanned, but won't perform an actual scan. -PS which will perform a TCP SYN scan by sending a SYN packet to the destination host, by default, on port 80. If an RST is received, it indicates a closed port. No reply indicates a filtered port, and an ACK response indicates an open port. Similarly, <code>-PA</code> (ACK ping), <code>-PR</code> (ARP ping), <code>-PU</code> (UDP Ping) are also available. While scanning internal networks, in particular, the <code>-n</code> option that disables the DNS resolution of IP addresses, may come in handy.